

Lecture 3: Understanding Risk

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Overview

This lecture studies how to measure risk and assess its impact on financial instrument prices and demand.

Key insights:

- Risk measurement tools originated from analyzing games of chance
- Risk cannot be eliminated, but can be effectively managed
- Risk creates opportunities - investors are compensated for bearing risk
- Measuring risk is essential for pricing and transferring risk

The Road Ahead

1. Defining Risk
2. Measuring Risk
3. Risk-Return Tradeoff
4. Sources of Risk
5. Reducing Risk through Diversification

Defining Risk

Risk is a measure of uncertainty about the future payoff to an investment, assessed over some time horizon and relative to a benchmark.

Remarks:

- Risk is a measure that can be quantified. Uncertainties that are not quantifiable cannot be priced
- Risk arises from uncertainty about the future
- Risk has to do with the future payoff of an investment, which is unknown
- This definition of risk refers to an investment or group of investments
- Risk must be assessed over some time horizon
- Risk must be assessed relative to a benchmark rather than in isolation

Measuring Risk

Probability is a measure, expressed as a number between zero and one, of the likelihood that an event will occur. We will use some elementary tools from probability theory to measure risk.

Example: If we flip a fair coin, the probabilities that it will come down head or tail are both one-half.

Key principle: The probabilities of all the possibilities are always non-negative and sum up to one.

We can now compute the **expected value** or mean of an investment with more than one possible returns or **payoffs**.

Case 1: Two Possible Payoffs

Suppose that for \$1,000 we can purchase a stock whose value is equally likely to fall to \$700 or rise to \$1,400.

If we had purchased this stock one million times, about half million times the stock would pay off \$700 and the other half million times it would pay off \$1,400.

Average payoff from one million investments:

$$\begin{aligned}\underbrace{\$1,050}_{\text{EV}} &= \frac{500,000 \times \$700 + 500,000 \times \$1,400}{1,000,000} \\ &= \frac{1}{2} \times \$700 + \frac{1}{2} \times \$1,400\end{aligned}$$

This is also the **expected value (EV)** of the stock. It shows that the expected value is the probability-weighted sum of the possible payoffs.

Case 2: Four Possible Payoffs

Suppose that the stock now can yield four possible payoffs:

- \$100 (probability 0.1)
- \$700 (probability 0.4)
- \$1,400 (probability 0.4)
- \$2,000 (probability 0.1)

Expected value of the stock:

$$\$1,050 = 0.1 \times \$100 + 0.4 \times \$700 + 0.4 \times \$1,400 + 0.1 \times \$2,000$$

Note that $0.1 + 0.4 + 0.4 + 0.1 = 1$.

Key observation: The two \$1,000 investments both yield an expected return of \$50, or an expected return rate of 5%, but they carry different levels of risk.

Measures of Risk

The two examples suggest that we measure risk by quantifying the spread among an investment's possible outcomes.

Risk-free asset: An investment whose future value is known with certainty (i.e. with probability one) and whose return is the risk-free rate of return.

Two measures of risk:

1. Variance and Standard Deviation
2. Value at Risk (VaR)

Variance and Standard Deviation

Variance (Var) is defined as the average of the squared deviations of the possible outcomes from their expected value, weighted by their probabilities.

For Case 1:

$$\underbrace{122,500 \text{ (dollars}^2\text{)}}_{\text{Var}} = \frac{1}{2}(\$1,400 - \$1,050)^2 + \frac{1}{2}(\$700 - \$1,050)^2$$

Standard deviation (s.d.) is simply the (positive) square root of the variance:

$$\text{s.d.} = \sqrt{\text{Var}} = \sqrt{122,500 \text{ (dollars}^2\text{)}} = \$350$$

This is more useful than the variance because it is measured in the same unit as the payoffs.

Comparing Standard Deviations

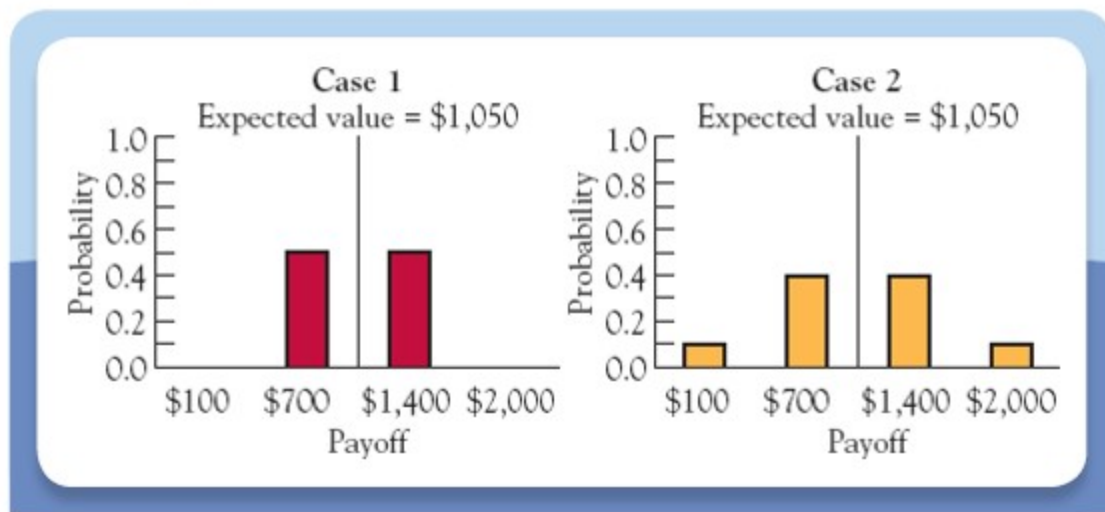
For Case 2: s.d. = \$528. Because the two investments have the same expected value, most people would prefer the first.

Key principle: The greater the standard deviation, the higher the risk.

Standard deviation measures the spread of all possible outcomes around the expected value.

Figure 5.1

Investing \$1,000



Value at Risk

Value at risk (VaR) is the worst possible loss over a specific time horizon, at a given probability.

It measures the riskiness of the worst outcome rather than the spread of all outcomes measured by standard deviation.

Comparison:

- Case 1: VaR is \$300 with probability 0.5
- Case 2: VaR is \$900 with probability 0.1

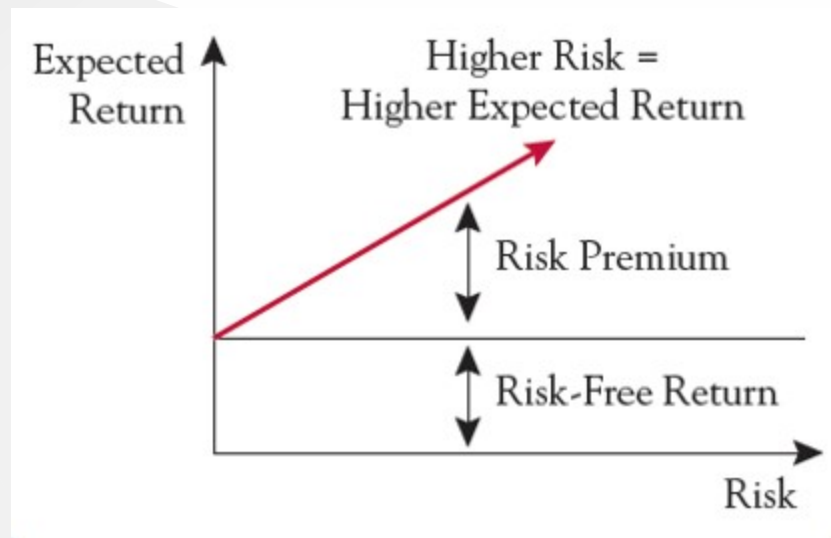
VaR focuses on downside risk rather than overall variability.

Risk-Return Tradeoff

Investor types: Risk-averse investors prefer certain returns over risky ones with the same expected return, while risk-neutral investors only care about expected return.

Key insights:

- Risky investments must offer an extra return, called **risk premium**, because investors require compensation for taking risk
- The riskier an investment, the higher the risk premium and expected return



Sources of Risk

We can classify all risks into one of two groups:

Idiosyncratic risks (or unique risks):

- Those affecting a small number of people but no one else
- Example: The risk that Ford's stock price may go up or down is idiosyncratic because it only affects Ford and its shareholders

Systematic risks (or economy-wide risks):

- Those affecting everyone
- Example: Macroeconomic factors, such as swings in consumer and business confidence, are sources of systematic risks that affect all firms and individuals

Reducing Risk through Diversification

Risk can be reduced through diversification, the principle of holding more than one risk at a time.

Key insight: Holding several different investments can reduce the idiosyncratic risk an investor bears.

Two ways to diversify an investment:

1. Hedging risk
2. Spreading risk

Hedging Risk

Hedging is the strategy of reducing idiosyncratic risk by making two investments with opposing risks.

Example: Suppose oil prices have an equal chance of rising or falling.

Texaco (oil company):

- Oil prices rise → \$120 payoff per \$100 invested
- Oil prices fall → \$100 payoff per \$100 invested

GE (general company):

- Oil prices rise → \$100 payoff per \$100 invested
- Oil prices fall → \$120 payoff per \$100 invested

Increases in oil price are bad for most of the economy, but good for oil companies.

Strategies without Hedging

Strategy 1: Invest \$100 in GE

Expected payoff:

$$\$110 = \frac{1}{2} \times \$120 + \frac{1}{2} \times \$100$$

Standard deviation:

$$\$10 = \sqrt{\frac{1}{2} \times (\$120 - \$110)^2 + \frac{1}{2} \times (\$100 - \$110)^2}$$

Strategy 2: Invest \$100 in Texaco

Same EV and s.d. as Strategy 1

Strategy with Hedging

Strategy 3: Invest \$50 in GE and \$50 in Texaco

Expected payoff:

$$\$110 = \frac{1}{2} \times (\$60 + \$50) + \frac{1}{2} \times (\$60 + \$50)$$

Standard deviation:

$$\$0 = \sqrt{\frac{1}{2} \times (\$60 + \$50 - \$110)^2 + \frac{1}{2} \times (\$60 + \$50 - \$110)^2}$$

Strategy 3 guarantees a payoff of \$110, regardless of whether oil prices go up or down.

Key insight: Hedging has eliminated the risk entirely in this example.

Spreading Risk

Spreading is another strategy of reducing idiosyncratic risk by making several investments with unrelated payoffs.

Example: Suppose that GE and Microsoft's payoffs are independent of each other and both pay off either \$120 or \$100 with equal probability for each \$100 invested.

Strategy 1: Invest \$100 in GE, EV = \$110, s.d. = \$10

Strategy 2: Invest \$100 in Microsoft, same EV and s.d. as Strategy 1

Strategy 3: Invest \$50 in GE and \$50 in Microsoft

Expected payoff:

$$\$110 = \frac{1}{4} \times (\$60 + \$60) + \frac{1}{4} \times (\$60 + \$50) + \frac{1}{4} \times (\$50 + \$60) + \frac{1}{4} \times (\$50 + \$50)$$

Standard deviation:

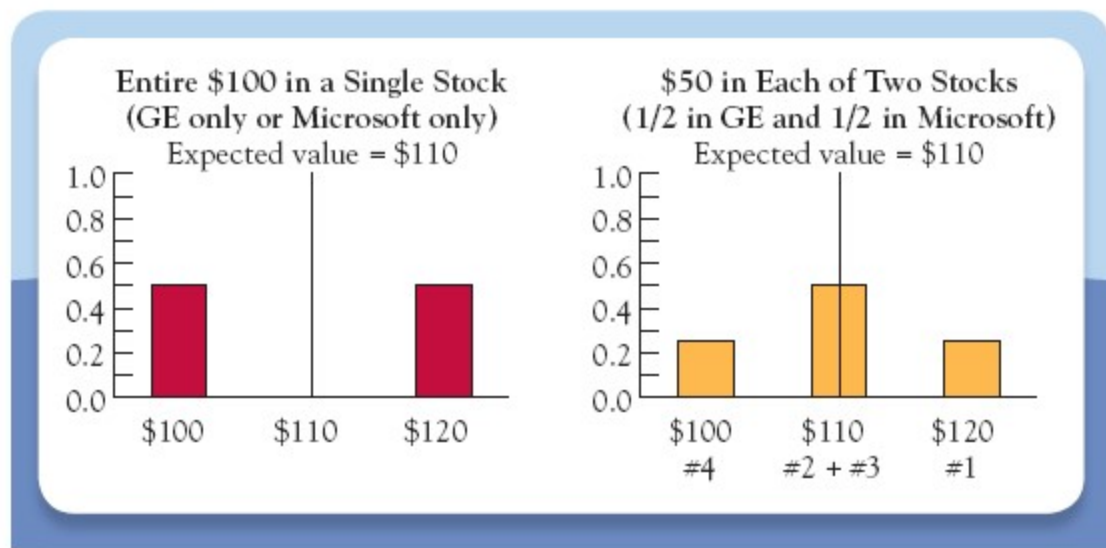
$$\$7.1 = \sqrt{\frac{1}{4} \times (\$60 + \$60 - \$110)^2 + \frac{1}{4} \times (\$50 + \$50 - \$110)^2}$$

Spreading Reduces Risk

By spreading the investment among independently risky investments, Strategy 3 lowers the spread of the outcomes and hence the risk.

When the investment is split between the two stocks, the payoff becomes \$110 or higher with probability 0.75 and \$100 with only probability 0.25.

Figure 5.5 Spreading Risk Payoffs from Two Investment Strategies



Math Appendix

We will show how diversification reduces risk in general.

Let (x, y) be the payoff pair to buying GE and Texaco stocks and p_i the probability associated with a particular outcome (x_i, y_i) .

The actual payoff on the investment is:

$$\text{actual payoff} = ax + by$$

where $0 \leq a, b \leq 1$ are the proportions of the investment made in GE and Texaco, respectively, and satisfy $a + b = 1$.

Risk Hedging

Variance formula:

$$\text{Var}(ax + by) = a^2\text{Var}(x) + b^2\text{Var}(y) + 2ab\text{Cov}(x, y)$$

where $\text{Cov}(x, y) = \sum_i p_i (x_i - \bar{x})(y_i - \bar{y})$ measures the extent to which two risky assets move together, and $\text{EV}(x) = \bar{x} = \sum_i p_i x_i$, $\text{EV}(y) = \bar{y} = \sum_i p_i y_i$.

GE/Texaco example: With $a = b = 1/2$:

$$\text{Var}(x) = \text{Var}(y) = 100, \quad \text{Cov}(x, y) = -100, \quad \text{Var}\left(\frac{1}{2}x + \frac{1}{2}y\right) = 0$$

Conclusion: The two stocks behave as hedges for each other.

Risk Spreading

Assume there are n independent investments $\{x^i\}_{i=1}^n$, each with the same expected payoff $\bar{x}$ and the same variance σ_x^2 .

If we hold $1/n$ of our portfolio in each stock, then the expected payoff is:

$$\text{EV} \left(\sum_{i=1}^n \frac{x^i}{n} \right) = \frac{1}{n} \sum_{i=1}^n \text{EV}(x^i) = \bar{x}$$

Since the payoff on each stock is independent of all the rest, all the covariances are zero and hence the variance is:

$$\text{Var} \left(\sum_{i=1}^n \frac{x^i}{n} \right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(x^i) = \frac{\sigma_x^2}{n}$$

Key result: The variance declines as n increases; when the value of n is extremely large, the variance is essentially zero.